TITLE: Creation of a Secure Multimodal Clinical and Radiographic Spine Surgery Dataset for Case-Control Research and Locally Hosted AI Development

Principal Investigator: Gregory Schwing, MD, PhD

Sponsor: David Edelman, MD

Co-investigators: Members of the Wayne State Open Spine Consortium

INTRODUCTION

Value, defined as healthcare outcomes relative to the cost spent to achieve them, is the cornerstone of modern healthcare reform. While national efforts aim to improve value-based care, the integration of multimodal artificial intelligence (AI) into preoperative planning and postoperative outcome prediction remains a frontier in quality improvement (QI) methodologies. This study aims to retrospectively assess clinical outcomes for consecutive patients who underwent spine surgery throughout the Detroit Medical Center (DMC) system.

Beyond traditional metrics, this project introduces an advanced computational framework. We will integrate clinical text (operative notes, radiology reports) with radiographic imaging (primarily CT scans) to develop a proprietary, multimodal Large Language Model (LLM). This "report-and-image-to-mask" system will assist in the automated identification of anatomical variants—such as Lumbosacral Transitional Vertebrae (LSTV)—and the prediction of surgical trajectories. By leveraging a diverse patient cohort, this study seeks to provide a high-quality benchmark for neurosurgical AI while objectively assessing and improving the value of care at our institution.

STUDY OBJECTIVES

The primary purpose of this study is to retrospectively assess quality metrics and develop a deep learning-ready multimodal database for the spine surgery service line at the Detroit Medical Center.

- **Primary Clinical Outcomes:** Compare length of stay (LOS), patient disposition, ICU admission, readmission rates, and in-hospital complications.
- **Primary Computational Objectives:** Train a private, locally hosted multimodal LLM to recognize complex spinal anatomy and provide clinical decision support, including the generation of synthetic planning imagery (e.g., hardware/screw placement) and educational visualizations for rare pathologies like spina bifida and scoliosis.

STUDY DESIGN

The design is a retrospective cohort study. To identify our dataset, we will request specific CPT and ICD/DRG codes and utilize departmental surgical logs to search the DMC electronic

medical record (EMR). To ensure the highest level of privacy and institutional ownership, **all deep learning and AI training will be conducted on the secured Wayne State University (WSU) high-performance computing grid.** No data will be exposed to public or online AI systems (e.g., ChatGPT).

Initially, records will be coded to allow for data linking; however, once the dataset is finalized for AI training and shared research, all 18 HIPAA identifiers will be removed. Radiographic images will be converted to NIfTI format to strip metadata. The resulting dataset will be stored in a secure WSU REDCap environment and the encrypted WSU Grid.

METHODS

Following IRB approval, data from the last 10 years will be collected. To ensure data integrity, 15% of the charts will be double-coded by independent reviewers. Automated pseudolabeling will be used for initial imaging segmentation, subsequently confirmed by supervised medical students.

Variables of interest include:

- **Patient Demographics & Baseline Health:** Standard anthropometric data and preoperative clinical status.
- **Surgical Details:** Procedural characteristics, instrumentation specifics, and intraoperative metrics.
- **Radiographic Variables:** CT/MRI parameters, identification of anatomical variants (e.g., LSTV), and AI-generated anatomical masks.
- **Clinical Outcomes:** Postoperative complications, resource utilization, and patient-reported outcome measures.

Statistical Analysis: Categorical data will be analyzed using chi-square analysis, and continuous data will be analyzed with independent t-tests or Mann-Whitney tests. Deep learning performance will be evaluated via Dice similarity coefficients, sensitivity, specificity, and AUC-ROC. Clinical utility of generative AI outputs will be assessed via expert surgeon grading. Statistical significance is defined as $p < 0.05$.

SUBJECT SELECTION

The DMC EMR (Cerner) and Surginet data will be queried.

- **Inclusion Criteria:** Adult patients (≥18 years) who underwent spine surgery at a participating DMC facility with available radiographic imaging and corresponding clinical text records.
- **Exclusion Criteria:** Patients lacking either imaging or text records, miscoded charts, or cases with data corruption preventing de-identification.

RISKS

As a retrospective review, risks are minimal. The primary risk is a potential breach of confidentiality. This is strictly mitigated by:

1. Using HIPAA-compliant REDCap for data extraction.
2. Computational de-identification of all imaging and text.
3. Storage and AI training isolated to the encrypted, non-public WSU computing grid.

BENEFITS

While there are no direct benefits to participants, there are significant indirect benefits to society. This research facilitates the creation of AI models that democratize subspecialist expertise, improve surgical planning through generative hardware simulations, and provide a global benchmark for neurosurgical research.

USE OF INFORMATION & REGULATORY CONSIDERATIONS

This study is conducted under the supervision of **Dr. Gregory Schwing as Principal Investigator.** Dr. Schwing's dual background (MD and PhD in Computer Science) and history of publishing case-control studies provide the necessary expertise to manage the high-performance computing requirements and clinical datamining of this protocol.

DATA COLLECTION, RETENTION, AND MONITORING

All abstracted data will be kept password-protected and encrypted. Patient identifiers will be kept in a master linking log separate from the research data, accessible only to the PI. Data will be retained for 7 years to ensure reproducibility. Final disposal will involve secure digital shredding on the WSU Grid.